Exercise for chapter 1 to 3

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$PV = RT$

1.2 The expression for the total derivative of *V* with respect to the dependent variables *P* and *T* is

$\ln V = \ln \frac{T}{P} + C$

$$dV = \left(\frac{\partial V}{\partial P}\right)_T dP + \left(\frac{\partial V}{\partial T}\right)_P dT$$

$\frac{PV}{T} = e^C$

$R = e^C$

Substitute the values of β_T and α obtained in Qualitative Problem 2 into this equation and integrate them to obtain the equation of state for an ideal gas.

$\beta_T = \frac{1}{P}$ $\quad \alpha = \frac{1}{T}$

$dV = -\frac{V}{P} dP + \frac{V}{T} dT$

$\left(\frac{\partial V}{\partial P}\right)_T = -\beta_T V = -\frac{V}{P}$

$\left(\frac{\partial V}{\partial T}\right)_P = \alpha V = \frac{V}{T}$

$\int \frac{dV}{V} = -\int \frac{dP}{P} + \int \frac{dT}{T}$

$\ln V = -\ln P + \ln T + C$

Solution to 1.2

According to Qualitative Problem 2,

$$\beta_T = \frac{1}{P},\ \alpha = \frac{1}{T}.$$

So $\left(\frac{\partial V}{\partial P}\right)_T = -V\beta_T = -\frac{V}{P},\ \left(\frac{\partial V}{\partial T}\right)_P = V\alpha = \frac{V}{T}.$

$$dV = \left(\frac{\partial V}{\partial P}\right)_T dP + \left(\frac{\partial V}{\partial T}\right)_P dT = -\frac{V}{P}dP + \frac{V}{T}dT.$$

Integrate $\frac{dV}{V} = -\frac{dP}{P} + \frac{dT}{T}$,

$\ln V = -\ln P + \ln T + \mathrm{C} = \ln\frac{T}{P} + \mathrm{C}$ (C is constant).

So, $PV = e^{\mathrm{C}} \cdot T$.

Take $R = e^{\mathrm{C}}$, so for an ideal gas, the equation of state is

$PV = RT$.

1.3 The pressure temperature phase diagram in Figure 1.4 has no two-phase areas (only two-phase curves), but the temperature composition diagram in Figure 1.5 does have two-phase areas. Explain.

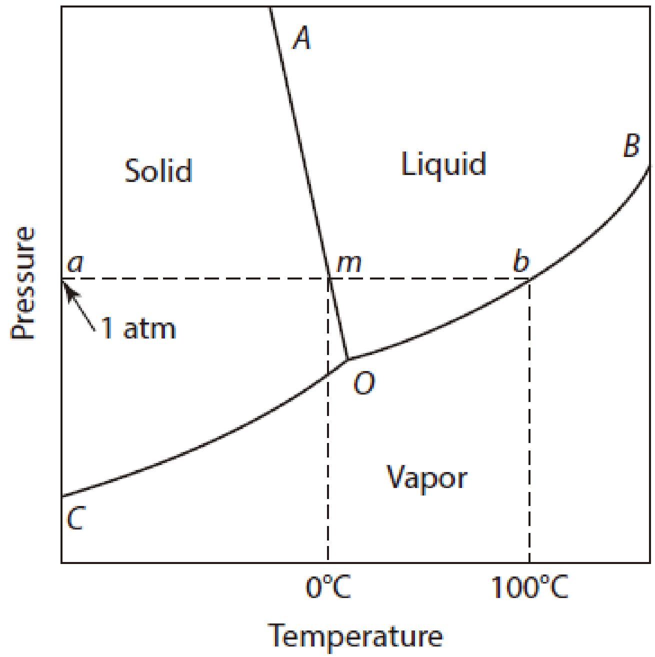

Figure 1.4 Schematic representation of part of the pressure–temperature equilibrium phase diagram for H_2O. The melting point is designated as *m* and the boiling point as *b*.

1.3 The pressure temperature phase diagram in Figure 1.4 has no two-phase areas (only two-phase curves), but the temperature composition diagram in Figure 1.5 does have two-phase areas. Explain.

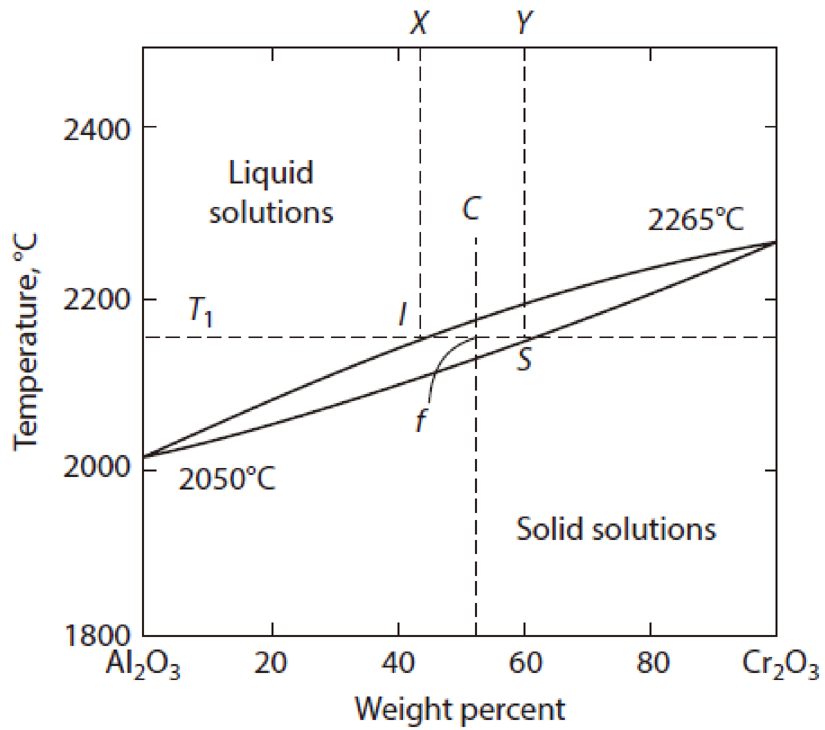

Figure 1.5 The temperature composition equilibrium phase diagram for the system Al_2O_3–Cr_2O_3 at a constant pressure of 1 atm.

Solution to 1.3

According to Phase Rule: F=C-P+2, where the degree of freedom F is a factor (such as temperature and pressure) that can be independently changed without changing the number of phases when the system is in equilibrium; C is the number of components in the system; P is the number of coexisting phases in the selected system.

a) In Figure 1.4, it has only one component, when there are two coexisting phases, F=1-2+2=1, so it has one degree of freedom, that is, it has only two-phase curves.

b) In Figure 1.5, it has two components, when there are two coexisting phases, F=2-2+2=2, so it has two degrees of freedom, that is it has two-phase areas.

Chapter 2

$T=300$ $V=15$

2.1 An monatomic ideal gas at 300 K has a volume of 15 liters at a pressure of 15 atm. Calculate:

$P=15$ $\frac{15\times 15}{10}$

a. The final volume of the system

b. The work done by the system $U=0.$ $Q=-W\int nRT\ln\frac{V_2}{V_1}$

c. The heat entering or leaving the system

d. The change in the internal energy $\Delta U=0.$

e. The change in the enthalpy when the gas undergoes $\Delta H=0.$

i. A reversible isothermal expansion to a pressure of 10 atm

ii. A reversible adiabatic expansion to a pressure of 10 atm

The constant-volume molar heat capacity of the gas, c_v, has the value 1.5 *R* .

Solution to 2.1

(i) Initial state : $p_1 = 15$ atm, $V_1 = 15$ liters, $T_1 = 300$ K

Process: isothermal reversible

Final state: $p_2 = 10$ atm, $V_2 =?$, $T_2 = 300$ K

a. Isothermal process, $p_1 V_1 = p_2 V_2$,

So $V_2 = \frac{p_1 V_1}{p_2} = \frac{15 \text{ atm} \cdot 15 \text{ liters}}{10 \text{ atm}} = 22.5$ liters

b. $p_1 V_1 = p_2 V_2 = nRT$,

$$w = \int_{V_1}^{V_2} PdV = \int_{V_1}^{V_2} \frac{nRT}{V} dV = nRT \ln V |_{V_1}^{V_2} = nRT \ln \frac{V_2}{V_1} = p_1 V_1 \ln \frac{V_2}{V_1}$$

$$= 15 \text{ atm} \times 15 \text{ liters} \ln \frac{22.5 \text{liters}}{15 \text{liters}}$$

Solution to 2.1

b. $w = 15 \text{ atm } 15 \text{ liters} \ln \frac{22.5\text{liters}}{15\text{liters}} =$

$$15 \times 101325 \text{ Pa} \times 15 \times 10^{-3} \text{ m}^3 \ln \frac{22.5}{15} = 9244 \text{ J}$$

For ideal gas, the internal energy is the function only of temperature. So:

c. $Q = w = 9244 \text{ J};$

d. $\Delta U = 0;$

e. $\Delta H = 0.$

(ii) Initial state : $p_1 = 15$ atm, $V_1 = 15$ liters, $T_1 = 300$ K

Process: Adiabatic reversible

Final state: $p_2 = 10$ atm, $V_2 =?$, $T_2 =?$

$$\frac{p_1}{p_2} = \left(\frac{V_2}{V_1}\right)^{\gamma}$$

a. $p_1V_1^{\gamma} = p_2V_2^{\gamma}$, for a monatomic ideal gas, $\gamma = \frac{5}{3}$.

So $V_2 = \left(\frac{p_1V_1^{\gamma}}{p_2}\right)^{\frac{1}{\gamma}} = \left(\frac{15\ \text{atm}15^{5/3}\ \text{liers}}{10\ \text{atm}}\right)^{\frac{3}{5}} = 19.13$ liters

b. $w = \int_{V_1}^{V_2} PdV = \int_{V_1}^{V_2} \left(\frac{p_1 V_1^{\gamma}}{V^{\gamma}}\right) dV \ = \frac{p_1 V_1^{\gamma}}{-\gamma+1} V^{-\gamma+1}\Big|_{V_1}^{V_2}$

$$= \frac{15 \text{ atm} 15^{5/3} \text{ liers}^{5/3}}{2/3} \left(19.13^{-2/3} - 15^{-2/3}\right) \text{liers}$$

$$= -1.5 \times 15 \times 101325 \times 15^{5/3} \times 10^{-3} \times \left(19.13^{-2/3} - 15^{-2/3}\right) \text{J} = 5119 \ \text{J}$$

$$w = -\Delta U = -nc_v \Delta T = -\frac{p_1 V_1}{RT_1} 1.5R\left(\frac{p_2 V_2}{nR} - T_1\right)$$

$$= -9.14 \times 1.5 \times 8.3145 \times (255.07 - 300) = 5121.65 \text{ J}$$

c. For adiabatic process, $Q = 0$

d. $\Delta U = -w = -5121.96\ \mathrm{J}$

e. $\Delta H = \Delta U + \Delta(PV)$

$$= -5121.96 + (10 \times 19.13 - 15 \times 15) \times 101325 \times 10^{-3}$$

$$= -8536.61\ \mathrm{J}$$

2.3 The initial state of a quantity of monatomic ideal gas is *P*=1 atm, *V*=1 liter, and *T*=373 K. The gas is isothermally expanded to a volume of 2 liters and is then cooled at constant pressure to the volume *V*. This volume is such that a reversible adiabatic compression to a pressure of 1 atm returns the system to its initial state. All of the changes of state are conducted reversibly. Calculate the value of *V* and the total work done on or by the gas.

$P_1=1 \quad V=1 \quad T=373 \qquad U=Q+w.$

$P_2=\frac{1}{2} \quad V_2=2 \quad T_2=373 \qquad ① w=nRT\ln\frac{V_2}{V_1}$

$P_3=P_2 \quad V_3= \quad T_3=\frac{373}{2} \qquad ② w=\int P dv=-\frac{1}{2}($

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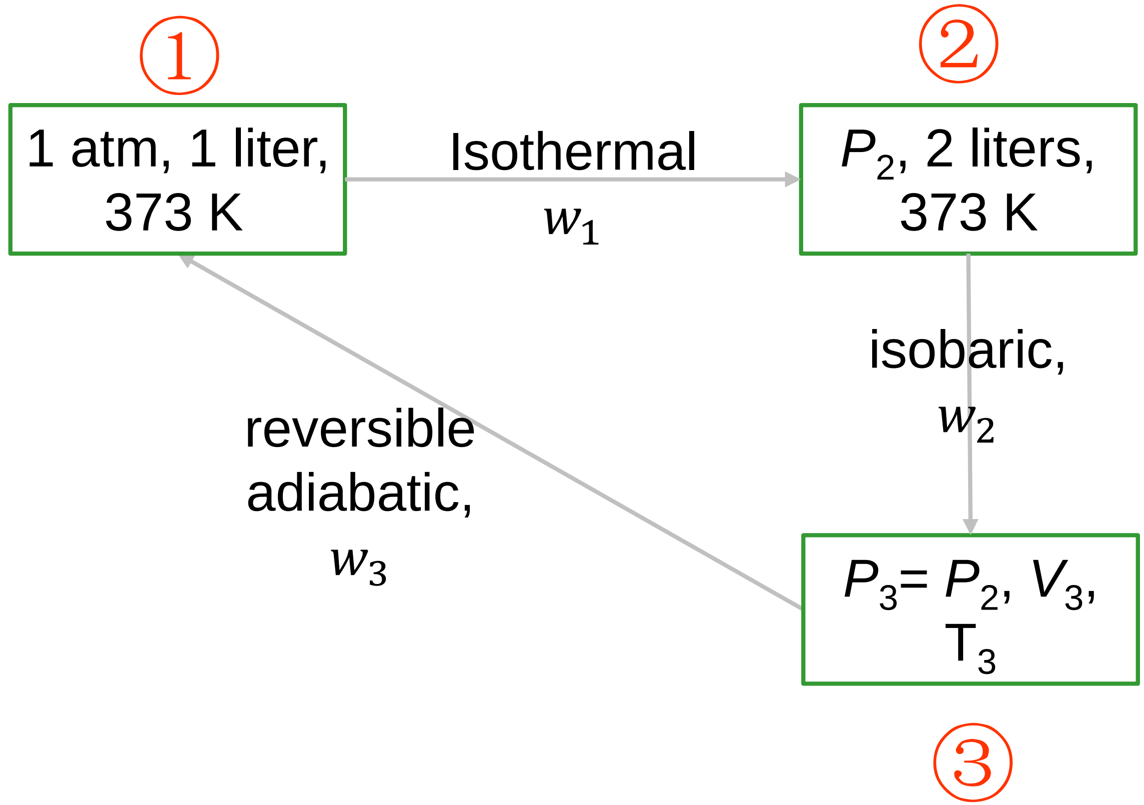
Solution to 2.3

The process is shown in the figure on the right.

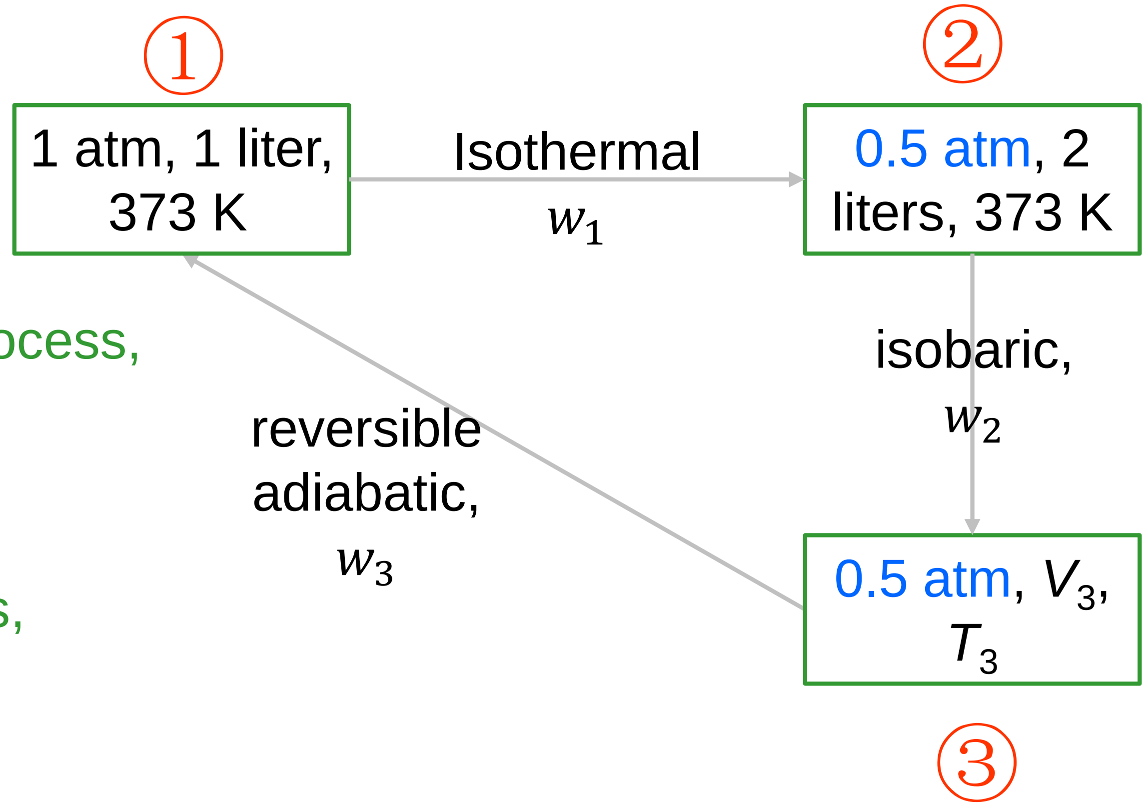


$$P_3 = P_2 = \frac{P_1 V_1}{V_2}$$

$$= \frac{1\ \text{atm} \times 1\ \text{liter}}{2\ \text{liters}}$$

$$= 0.5\ \text{atm}$$

Solution to 2.3



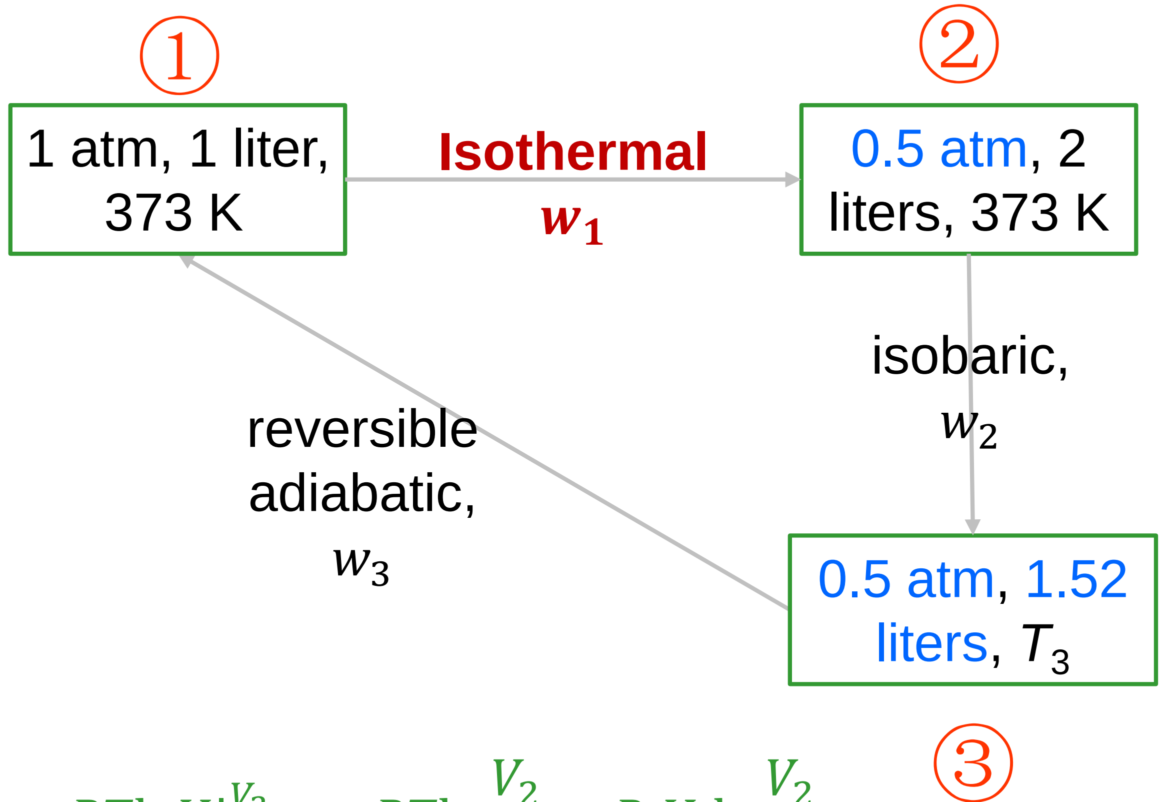
For reversible adiabatic process,

$$P_3 {V_3}^{\gamma} = P_1 V_1^{\gamma},$$

for an monatomic ideal gas,

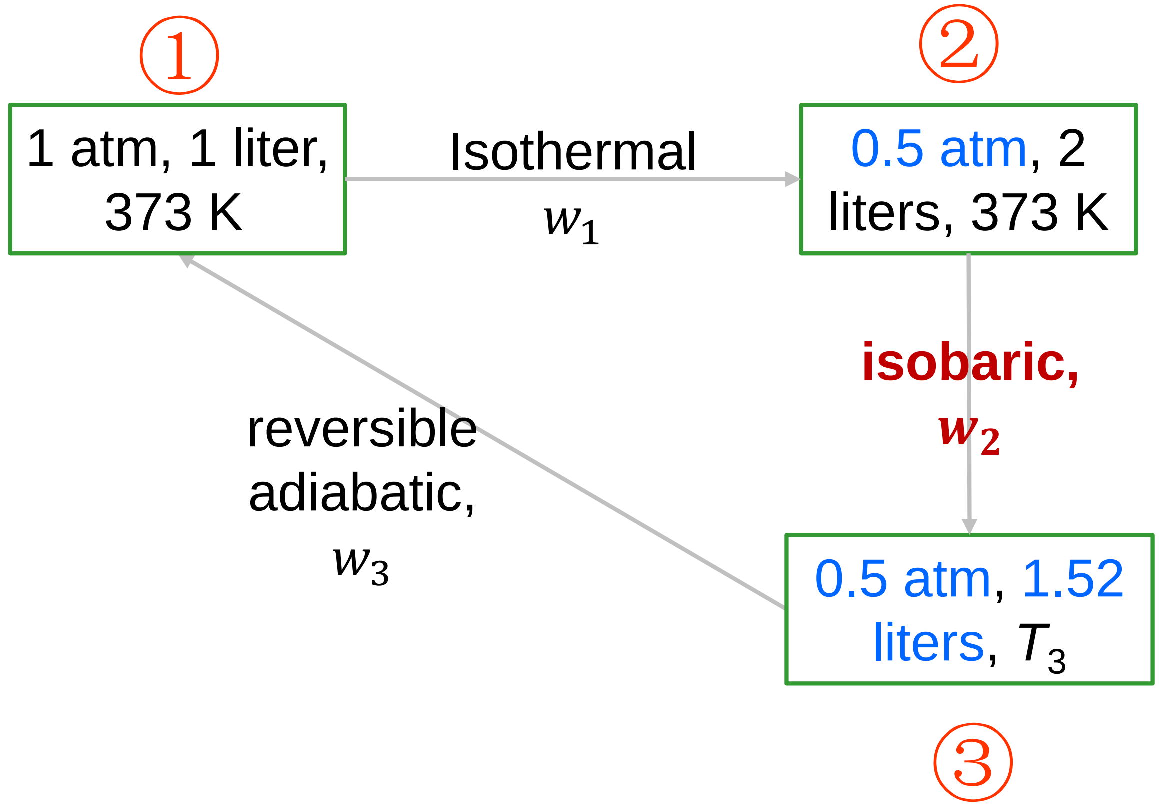
$$\gamma = \frac{5}{3}$$

$$V_3 = \left(\frac{P_1 V_1^{\gamma}}{P_3}\right)^{\frac{1}{\gamma}} = \left(\frac{1\ \text{atm} 1^{\gamma}\ \text{liter}^{\gamma}}{0.5\ \text{atm}}\right)^{\frac{1}{\gamma}} = 1.52\ \text{liters}$$



$$w_1 = \int_{V_1}^{V_2} PdV = \int_{V_1}^{V_2} \frac{nRT}{V} dV = nRT\ln V|_{V_1}^{V_2} = nRT\ln\frac{V_2}{V_1} = P_1V_1\ln\frac{V_2}{V_1}$$

$$= 1\ \mathrm{atm}\,1\ \mathrm{liter}\ln\frac{2}{1} = 1 \times 101325 \times 1 \times 10^{-3}\ln 2\ \mathrm{J} = 70.233\ \mathrm{J}$$

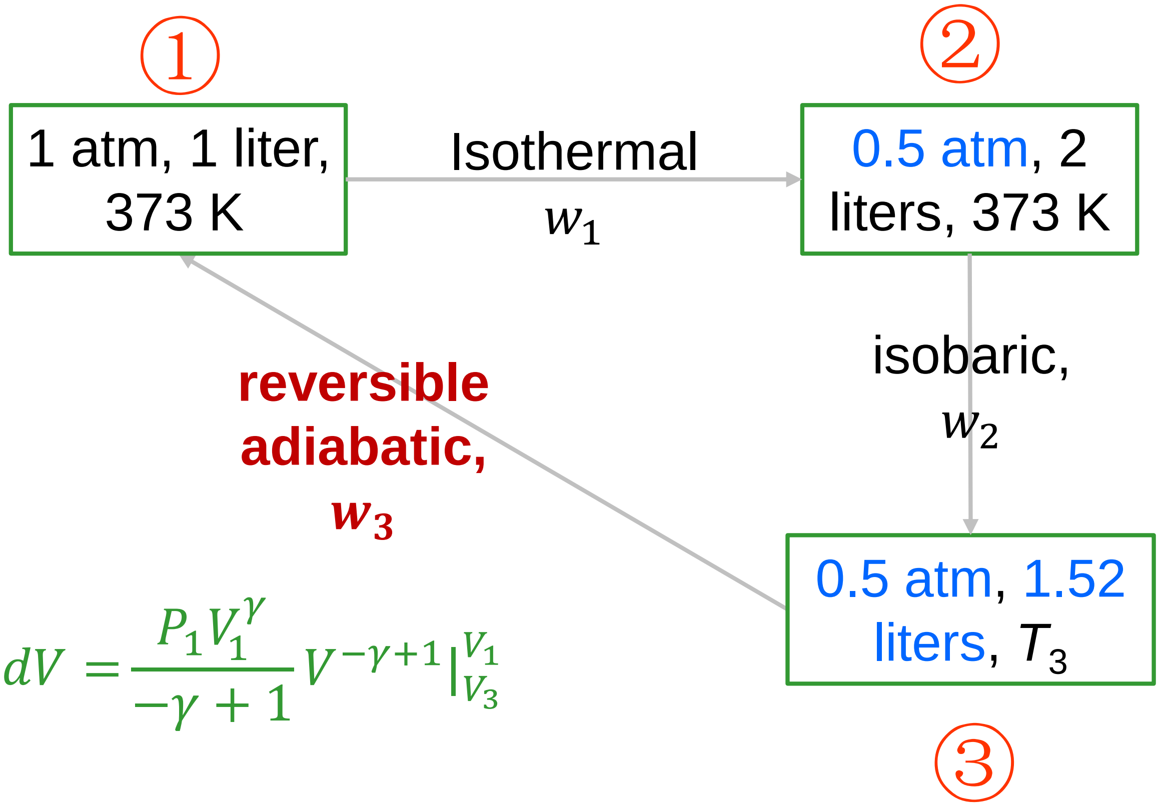


$$w_2 = P_2 \Delta V = P_2(V_3 - V_2)$$

$$= 0.5 \text{ atm} \times (1.52 - 2) \text{ liters} = 0.5 \times 101325 \times (1.52 - 2) \times 10^{-3} \text{ J}$$

$$= -24.318 \text{ J}$$

Chapter 2



$$w_3 = \int_{V_3}^{V_1} PdV = \int_{V_3}^{V_1} \left(\frac{P_1 V_1^{\gamma}}{V^{\gamma}}\right) dV = \frac{P_1 V_1^{\gamma}}{-\gamma + 1} V^{-\gamma+1}\Big|_{V_3}^{V_1}$$

$$= 1 \times 101325 \times (-1.5) \times \left(1 - 1.52^{-2/3}\right) \times 10^{-3}\ \mathrm{J} = -37.019\ \mathrm{J}$$

$$w = w_1 + w_2 + w_3 = (70.233 - 24.318 - 37.019)\ \mathrm{J} = 8.896\ \mathrm{J}$$

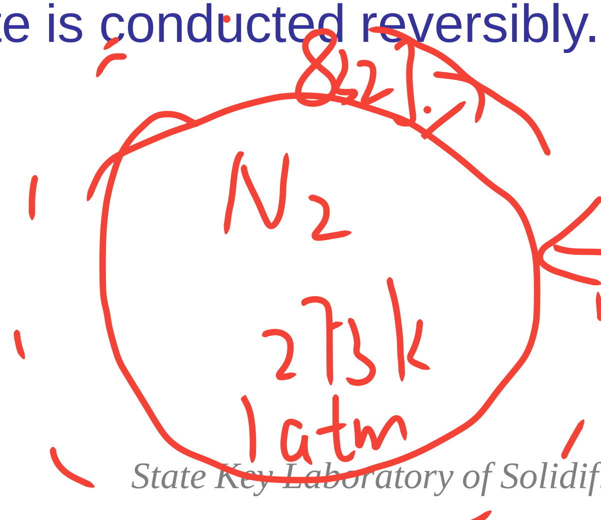
2.5 One mole of N_2 gas is contained at 273 K and a pressure of 1 atm. The addition of 3000 J of heat to the gas at constant pressure causes 832 J of work to be done during the expansion. Calculate

a. The final state of the gas

b. The values of ΔU and ΔH for the change of state

c. The values of c_v and c_p for N_2

Assume that nitrogen behaves as an ideal gas, and that the change of state is conducted reversibly.



$PV = nRT$

$V = \frac{RT}{P}$

$W = P(V_2 - V_1)$

$V_2 = W + V_1$

$T_2 = \frac{PV_2}{R}$

$\Delta U = Q - W$

$\Delta H = 3000$

P

Solution to 2.5

Initial state: 1 mole N_2, $P_1 = 1\ \text{atm}$, $V_1 = ??$, $T_1 = 273\ \text{K}$,

Process: isobaric reversible, ΔH=3000 J. W=832 J.

Final state: 1 mole N_2, $P_2 = 1\ \text{atm}$, $V_2 = ??$, $T_2 = ??$,

a. Constant pressure, $P_2 = P_1 = 1\ \text{atm}$, according to $P_1V_1 = nRT_1$,

so $V_1 = \frac{nRT_1}{P_1} = \frac{1\times 8.3144\times 273}{1\times 101325}\ \text{m}^3 = 0.0224\ \text{m}^3 = 22.4\ \text{liters}$

$$w = P_1 \Delta V = P(V_2 - V_1),$$

$$V_2 = \frac{w}{P_1} + V_1 = \left(\frac{832}{1\times 101325}\times 10^3 + 22.4\right)\text{liters} = 30.6\ \text{liters}$$

$$T_2 = \frac{P_2V_2}{nR} = \frac{1\times 101325\times 30.6\times 10^{-3}}{1\times 8.3144}\text{K} = 373\ \text{K}$$

Chapter 2

b. $\Delta H = q_P = 3000\ \mathrm{J}$

$\Delta U = q_P - w = (3000 - 832)\ \mathrm{J} = 2168\ \mathrm{J}$

c. $c_v = \frac{\Delta U}{n\Delta T} = \frac{2168}{373-273} \frac{\mathrm{J}}{\mathrm{mol \cdot K}} = 21.68\ \mathrm{J/mol \cdot K}$

$c_p = \frac{\Delta H}{n\Delta T} = \frac{3000}{373-273} \mathrm{J/mol \cdot K} = 30\ \mathrm{J/mol \cdot K}$

$C_v = \frac{\Delta U}{\Delta T}$

$C_p = \frac{\partial H}{\partial T}$

Chapter 2

2.7 One mole of an monatomic ideal gas at 25 °C and 1 atm undergoes the following reversibly conducted cycle:

a. An isothermal expansion to 0.5 atm, followed by

b. An isobaric expansion to 100 °C, followed by

c. An isothermal compression to 1 atm, followed by

d. An isobaric compression to 25 °C

The system then undergoes the following reversible cyclic process:

a. An isobaric expansion to 100 °C, followed by

b. A decrease in pressure at constant volume to the pressure *P* atm, followed by

c. An isobaric compression at *P* atm to 24.5 liters, followed by

d. An increase in pressure at constant volume to 1 atm

Calculate the value of ***P*** which makes the work done on the gas during the first cycle equal to the work done by the gas during the second cycle.

25, 1atm $\xrightarrow{T}$ 25, 0.5atm

$\downarrow P$

100°C, 0.5atm $\xleftarrow{T}$ 100°C, 1atm

$\uparrow P$

P_1 & $\frac{PV}{298} = R$

$W_1 = nRT \ln \frac{P_1}{P_2}$

$W_2 = \int P dV$

273, 25, 298R

25, 1atm $\xrightarrow{P}$ 100, 1atm

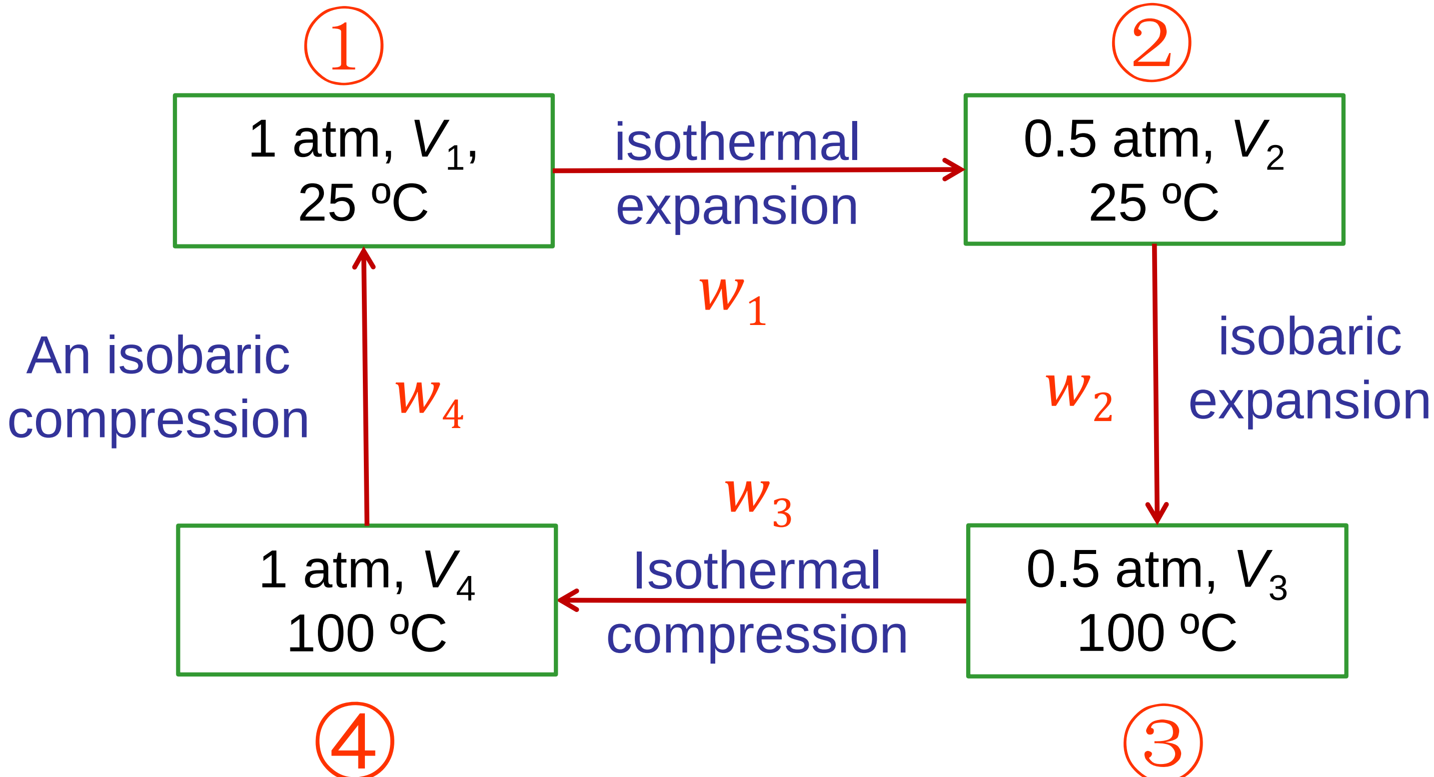
V↑ P 24.5V P ↓V P

Solution to 2.7

One mole of an monatomic ideal gas

The first reversible cyclic process

a. An isothermal expansion to 0.5 atm,
b. An isobaric expansion to 100 °C,
c. An isothermal compression to 1 atm,
d. An isobaric compression to 25 °C



$$\begin{array}{ccc} \text{①} & & \text{②} \\ \boxed{\begin{array}{c}1\text{ atm}, V_1,\\ 25\ ^{o}\text{C}\end{array}} & \xrightarrow[\text{isothermal expansion}]{w_1} & \boxed{\begin{array}{c}0.5\text{ atm}, V_2\\ 25\ ^{o}\text{C}\end{array}} \\ & & \Big\downarrow \text{isobaric expansion} \quad w_2 \\ & & \boxed{\begin{array}{c}0.5\text{ atm}, V_3\\ 100\ ^{o}\text{C}\end{array}} \\ & & \text{③} \end{array}$$

$$w_1 = \int_{V_1}^{V_2} PdV = \int_{V_1}^{V_2} \frac{nRT_1}{V} dV = nRT_1 \ln V |_{V_1}^{V_2}$$

$$= nRT_1 \ln \frac{V_2}{V_1} = nRT_1 \ln \frac{P_1}{P_2}$$

$$= 1 \times 8.3144 \times 298 \ln \frac{1}{0.5} \text{ J} = 1717.404 \text{ J}$$

$$w_2 = P_2 \Delta V = 0.5 atm \left(\frac{nRT_3}{P_3} - \frac{nRT_2}{P_2} \right)$$

$$= 0.5 \left(\frac{8.3144 \times 373}{0.5} - \frac{8.3144 \times 298}{0.5} \right) \text{ J} = 623.58 \text{ J}$$

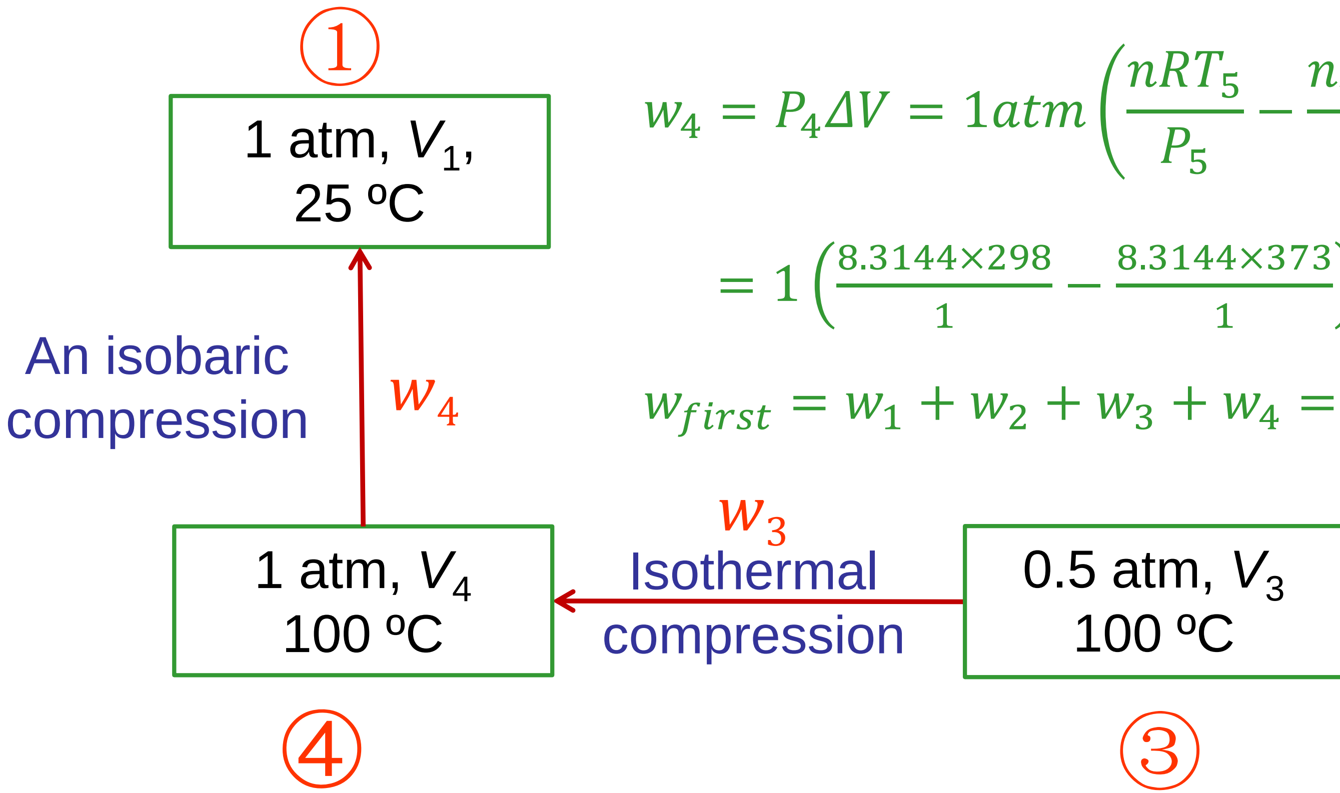
$$w_3 = \int_{V_3}^{V_4} PdV = \int_{V_3}^{V_4} \frac{nRT_3}{V} dV = nRT_3 \ln V|_{V_3}^{V_4} = nRT_3 \ln\frac{V_4}{V_3} = nRT_3 \ln\frac{P_3}{P_4}$$

$$= 1 \times 8.3144 \times 373 \ln\frac{0.5}{1}\ \mathrm{J} = -2149.637\ \mathrm{J}$$

$$w_4 = P_4 \Delta V = 1atm\left(\frac{nRT_5}{P_5} - \frac{nRT_4}{P_4}\right)$$

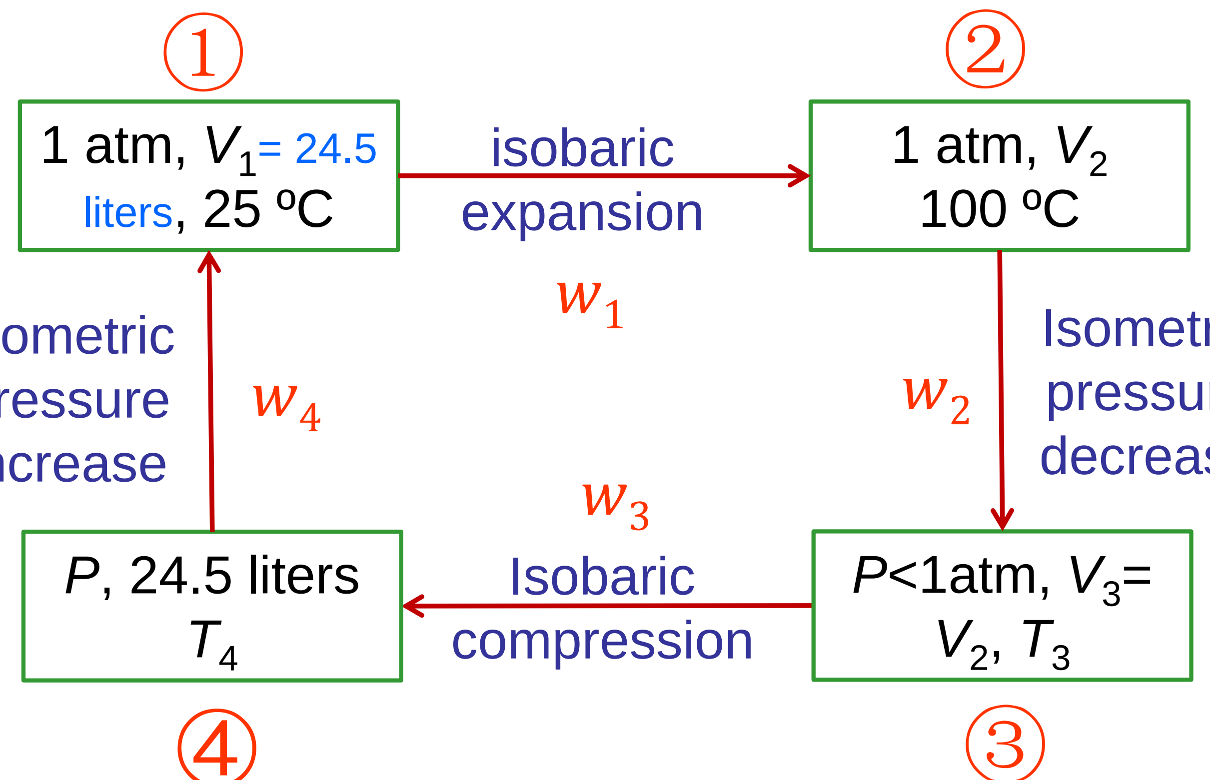
$$= 1\left(\frac{8.3144 \times 298}{1} - \frac{8.3144 \times 373}{1}\right)\ \mathrm{J} = -623.58\ \mathrm{J}$$

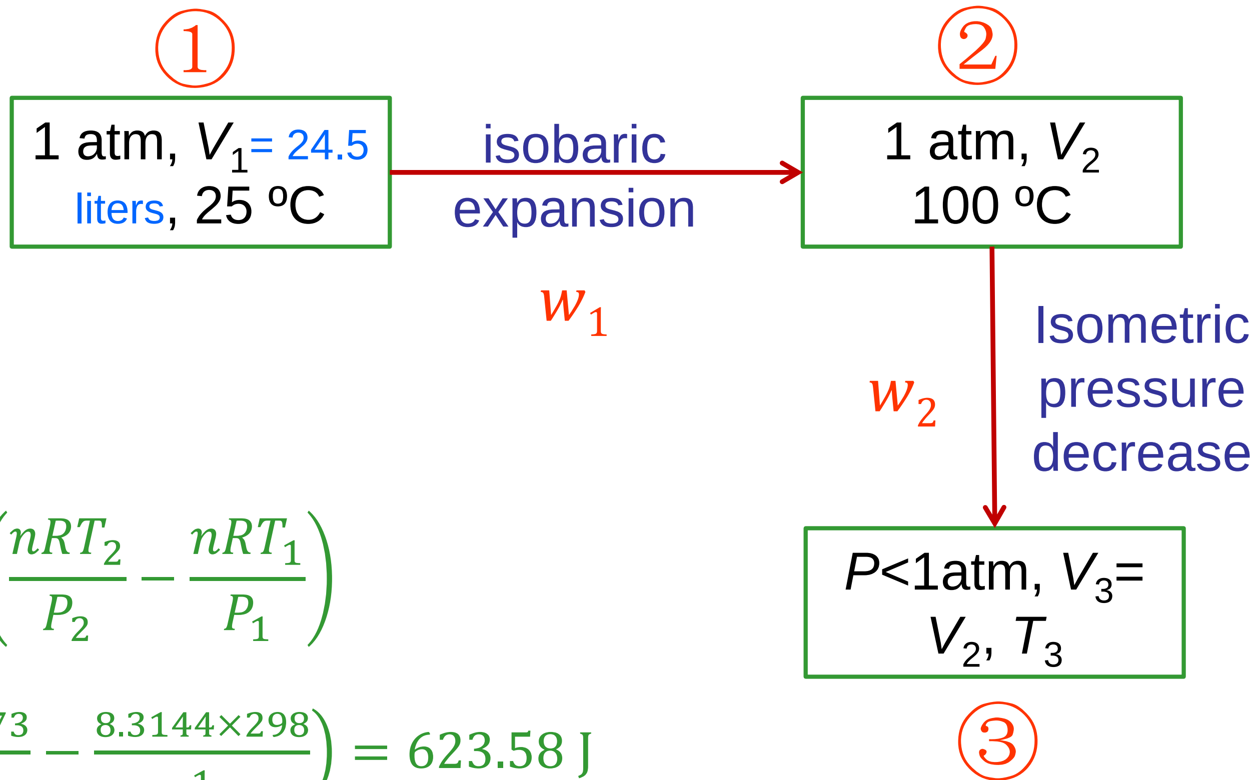
$$w_{first} = w_1 + w_2 + w_3 + w_4 = -432.233\ \mathrm{J}$$



The second reversible cyclic process

a. An isobaric expansion to 100 °C,
b. A decrease in pressure at constant volume to the pressure P atm,
c. An isobaric compression at P atm to 24.5 liters
d. An increase in pressure at constant volume to 1 atm





$$w_1 = P_1 \Delta V = 1\ \text{atm}\left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1}\right)$$

$$= 1\ \text{atm}\left(\frac{8.3144\times 373}{1} - \frac{8.3144\times 298}{1}\right) = 623.58\ \text{J}$$

$$w_2 = 0$$

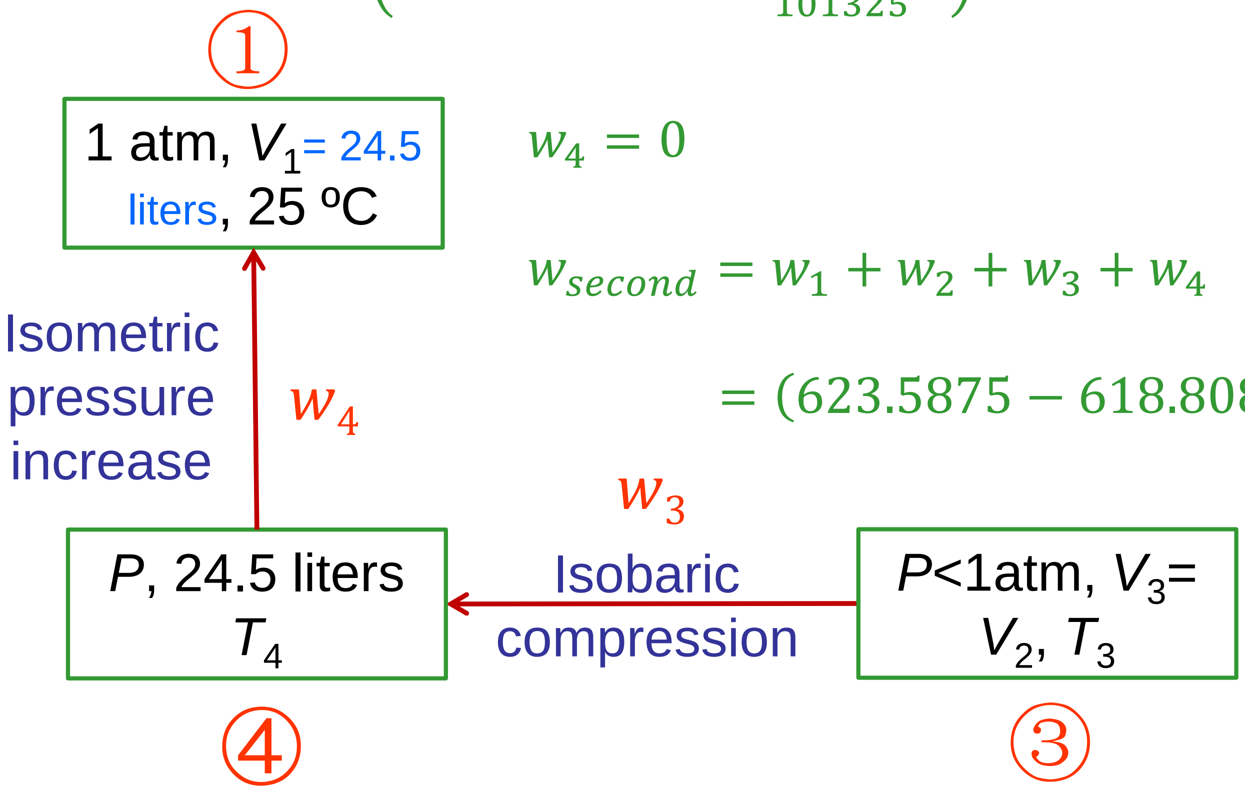
$$w_3 = P_3 \Delta V = P\text{atm}\left(V_4 - \frac{nRT_2}{P_2}\right) = P\text{ atm} \times (24.5\text{ liters} - \frac{8.3144\times 373}{1\text{ atm}})\text{ J}$$

$$= P \times 101325\left(24.5\times 10^{-3} - \frac{8.3144\times 373}{101325}\right)\text{ J} = -618.8087P\text{ J}$$

$$w_4 = 0$$

$$w_{second} = w_1 + w_2 + w_3 + w_4$$

$$= (623.5875 - 618.8087P)\text{ J}$$



Because the work done on the gas during the first cycle equal to the work done by the gas during the second cycle,

$$w_{second} = -w_{first}$$

That is, $(623.5875 - 618.8087P)\ \mathrm{J} = 432.233\ \mathrm{J}$

So $P = (623.5875 - 432.233) \div 618.8087 = 0.3\ \mathrm{atm}$

3.1 The initial state of 1 mole of a monatomic ideal gas is P = 10 atm and T = 300 K. Calculate the change in the entropy of the gas for

a. An isothermal decrease in the pressure to 5 atm

b. A reversible adiabatic expansion to a pressure of 5 atm

c. A constant-volume decrease in the pressure to 5 atm

$V=0$

a. $PV = nRT$

$V = \frac{R300}{10}$

$S = \int \frac{q}{T} d$

b. $\Delta S = 0$

c. $V = \frac{R300}{10}$

$\Delta S = \int \frac{q}{T} dV$

10atm, $\frac{R300}{10}$, T 300K → 5atm, $\frac{R300}{10}$, T=150

$\frac{PV}{nT} = \frac{PV}{T}$ $\frac{10}{300}$ $\frac{5}{150}$

$\int \frac{nC_v dT}{T} = nR1.5\ln\frac{1}{2}$

Solution to 3.1

Substance: 1 mole of a monatomic ideal gas

Initial state: P_1 = 10 atm, T_1 = 300 K, V_1= ??

Process: An isothermal decrease in the pressure

Final state: P_2= 5 atm, T_2 = 300 K, V_2= ??

a. Isothermal process, $\Delta U = 0$,

so, $q = w = \int_{V_1}^{V_2} PdV = nRT\ln\frac{V_2}{V_1}$

$$\Delta S = \frac{q}{T} = nR\ln\frac{V_2}{V_1} = nR\ln\frac{P_1}{P_2} = 1 \times 8.3144 \times \ln\frac{10}{5}\ \mathrm{J/K} = 5.76\ \mathrm{J/K}$$

Solution to 3.1

Substance: 1 mole of a monatomic ideal gas

Initial state: P_1 = 10 atm, T_1 = 300 K, V_1= ??

Process: Reversible adiabatic expansion

Final state: P_2= 5 atm, T_2 = ??, V_2= ??

b. Reversible adiabatic process, $q = 0$, $\Delta S = 0$.

Solution to 3.1

Substance: 1 mole of a monatomic ideal gas

Initial state: P_1 = 10 atm, T_1 = 300 K, V_1= ??

Process: Isometric decrease in the pressure

Final state: P_2= 5 atm, T_2 = ??, V_2= ??

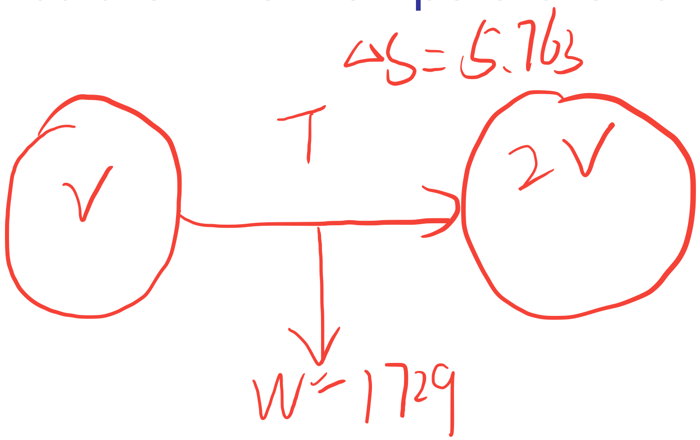
c. Constant-volume process, $\Delta V = 0$, so $w = 0$,

$$q = \Delta U = \int nc_v dT$$

$$\Delta S = \int_{T_1}^{T_2} \frac{nc_v}{T} dT = nc_v \ln\frac{T_2}{T_1} = nc_v \ln\frac{P_2}{P_1}$$

$$= 1 \times 1.5 \times 8.3144 \ln\frac{5}{10}\ \mathrm{J/K} = -8.64\ \mathrm{J/K}$$

3.3 One mole of a monatomic ideal gas undergoes a reversible expansion at constant pressure, during which the entropy of the gas increases by 14.41 J/K and the gas absorbs 6236 J of thermal energy. Calculate the initial and final temperatures of the gas. One mole of a second monatomic ideal gas undergoes a reversible isothermal expansion, during which it doubles its volume, performs 1729 J of work, and increases its entropy by 5.763 J/K. Calculate the temperature at which the expansion was conducted.



$\Delta S = \frac{W}{T} = \frac{1729}{x \cdot T}$ $T = 300K$

Solution to 3.3

(i) Reversible expansion at constant pressure,

Substance: One mole of a monatomic ideal gas

Process: The entropy of the gas increases by 14.41 J/K

The gas absorbs 6236 J of thermal energy

$$\Delta S = nc_p \ln\frac{T_2}{T_1} = 1 \times 2.5 \times 8.3144 \times \ln\frac{T_2}{T_1}\ \mathrm{J/K} = 14.41\ \mathrm{J/K}$$

Solve it, $\frac{T_2}{T_1} = 2$ (1)

$$q_P = nc_p(T_2 - T_1) = 1 \times 2.5 \times 8.3144 \times (T_2 - T_1)\ \mathrm{J} = 6236\ \mathrm{J}$$

Solve it, $(T_2 - T_1) = 300\ \mathrm{K}$ (2)

Combining (1) and (2), $T_1 = 300\ \mathrm{K}$, $T_2 = 600\ \mathrm{K}$.

Solution to 3.3

(ii) Reversible isothermal expansion,

Substance: One mole of a monatomic ideal gas

Process:

it doubles its volume

performs 1729 J of work

increases its entropy by 5.763 J/K

$$\Delta U = 0,$$

So $$w = q = T\Delta S$$

$$T = \frac{w}{\Delta S} = \frac{1729}{5.763}\ \mathrm{K} = 300\mathrm{K}$$

3.5 One mole of copper at a uniform temperature of 0 °C is placed in thermal contact with a second mole of copper which, initially, is at a uniform temperature of 100 °C. Calculate the temperature of the 2-mole system, which is contained in an adiabatic enclosure, when thermal equilibrium is attained.

Why is the common uniform temperature not exactly 50 °C?

How much thermal energy is transferred, and how much entropy is produced by the transfer?

The constant-pressure molar heat capacity of solid copper varies with temperature as c_p = 22.64 + 6.28 $\times$ 10^{-3} T J/mole·K

Solution to 3.5

According to the item, $T_1 = 273.15\text{K}$, $T_2 = 373.15\text{K}$,

$$q = \int nc_p dT = \int (22.64 + 6.28 \times 10^{-3} T) dT$$

$$q_{total} = \int_{T_1}^{T_e} nc_p \, dT + \int_{T_2}^{T_\mathrm{e}} nc_p \, dT$$

$$= [22.64T + 3.14 \times 10^{-3} T^2]_{T_1}^{T_e} + [22.64T + 3.14 \times 10^{-3} T^2]_{T_2}^{T_\mathrm{e}}$$

$= 0$ (adiabatic enclosure system)

Solve this, $T_e = 323.46\ \text{K}$

$$q_{trans} = \int_{T_1}^{T_e} nc_p\, dT = [22.64T + 3.14 \times 10^{-3}T^2]_{T_1}^{T_e} = 1236.92\ \mathrm{J}$$

$$\Delta S_{irr} = \int_{T_1}^{T_e} \frac{nc_p}{T} dT + \int_{T_2}^{T_e} \frac{nc_p}{T} dT$$

$$= [22.64\ln T + 6.28 \times 10^{-3}T]_{T_1}^{T_e} + [22.64\ln T + 6.28 \times 10^{-3}T]_{T_2}^{T_e}$$

$$= 0.619\ \mathrm{J/K}$$

Solution to 3.5

The final temperature is $T_e = 323.46\ \mathrm{K}$, which is greater than 323 K because the heat capacity increases with increasing temperature.

Thus the decrease in temperature caused by withdrawing heat *q* from hot copper is less than the increase in temperature caused by adding heat *q* to cold copper.

Review of chapter 1 to 3

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- ✓ Heat is identified as the process in which energy is transferred from one region to another down a temperature gradient.
- ✓ Work is identified as the process in which energy is transferred from one region to another not due to a temperature gradient.

- ✓ The system is that part of the universe we wish to investigate in detail.
- ✓ The surroundings is that part of the universe outside the system which may interact with it by exchanging energy or matter.
- ✓ The simple thermodynamic systems, is the case in which the surroundings interacts with the system only via pressure and temperature changes. The composition remains constant in simple systems

- ✓ *Isolated systems*: In these systems, no work is done on or by the system. In addition, energy or matter may not enter or leave it. Thus, the energy of these systems remains constant, as does the overall composition. Isolated systems are therefore unaffected by changes in the surroundings.
- ✓ *Closed systems*: These are systems which may receive (or give off) energy from (or to) the surroundings. The boundaries are called diathermal; that is, they allow thermal energy to transfer through them into or out of the system.
- ✓ *Open systems*: These are systems which can exchange both energy and matter with the surroundings. Neither the energy nor the composition of these systems need remain constant.

The boundaries or walls of the system are classified as follows:

- ✓ *Adiabatic:* No thermal energy can pass through.
- ✓ *Diathermal:* Thermal energy can pass through.
- ✓ *Permeable:* Matter can pass through.
- ✓ *Impermeable*: Matter cannot pass through.
- ✓ *Semipermeable*: Some components are able to pass through, while others are not.

- ✓ If it was possible to know the masses, velocities, positions, and all modes of motion of all of the constituent particles in a system, this mass of knowledge would serve to describe the *microscopic state* of the system, which, in turn, would determine all of the properties of the system.
- ✓ In the absence of such detailed knowledge as is required to determine the microscopic state of the system, thermodynamics begins with a consideration of the properties of the system which, when determined, define the *macroscopic state* of the system; i.e., when all of the properties are fixed then the macroscopic state of the system is fixed.

Chapter 1

- ✓ *independent variables*
- ✓ *dependent variables*
- ✓ *thermodynamic variables*
- ✓ *Extrinsic properties*
- ✓ *equation of state*
- ✓ The isothermal compressibility with dimensions of P^{-1}.

$$\beta_T = -\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T$$

- ✓ The coefficient of thermal expansion with dimensions of T^{-1}.

$$\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P$$

- ✓ Absolute zero of temperature
- ✓ the ideal gas temperature scale

$$1 \quad \text{atm} = 101325 \quad \text{newtons/meter}^2$$

$$1 \quad \text{liter} \cdot \text{atm} = 101.325 \quad \text{newton} \cdot \text{meters} = 101.325 \quad \text{joules}$$

$$R = 0.082057 \quad \text{liter} \cdot \text{atm/degree} \cdot \text{mole} = 8.314 \quad \text{joules/degree} \cdot \text{mole}$$

- ✓ Extensive properties have values which depend on the size of the system, and the values of intensive properties are independent of the size of the system.
- ✓ Volume is an extensive property, and temperature and pressure are intensive properties.

- A phase is defined as being a finite volume in the physical system within which the properties are uniformly constant, i.e., do not experience any abrupt change in passing from one point in the volume to another. Within any of the one phase areas in the phase diagram, the system is said to be homogeneous.

- The system is heterogeneous when it contains two or more phases, e.g., coexisting ice and liquid water (on the line AO) is a heterogeneous system comprising two phases, and the phase boundary between the ice and the liquid water is that very thin region across which the density changes abruptly from the value for homogeneous ice to the higher value for liquid water.

- ✓ The First Law not only states that the energy of the universe is conserved but it also posits that the various forms of energy (e.g., thermal, electrical, magnetic, and mechanical) can be converted into other forms of energy. When first delineated, it was the fact that thermal energy (heat) could be converted to mechanical work that was of special interest (heat engines). This law also defines an important extensive thermodynamic state function called the internal energy, U, of the system under investigation.

- ✓ Although the Second Law is the one that often gets the most attention in popular discussions of science. This law allows us to make important predictions of the direction in which a system will evolve with time during spontaneous processes, if other important caveats are taken into account.

- ✓ The Second Law introduces another important extensive thermodynamic state function called entropy, S . A short version of the Second Law is that the entropy of the universe never decreases.

- ✓ The Third Law states that when a system which is in complete internal equilibrium approaches the absolute zero in temperature, all of the aspects of its entropy approach zero.
- ✓ Sometimes, the Third Law is stated as follows: a system can never be taken to the absolute zero in temperature. This is also called the unattainability principle .

- ✓ a *mechanical equivalent of heat*
- ✓ In describing work, it is conventional to assign a negative value to work done on a body and a positive value to work done by a body
- ✓ In describing heat changes it is conventional to assign a negative value to heat which flows out of a body (an exothermic process) and a positive value to heat which flows into a body (an endothermic process).

$$\Delta U = U_B - U_A = q - w$$

$$dU = \delta q - \delta w$$

The first law of thermodynamics

- ✓ The use of the symbol "*d*" indicates a differential element of a state function or state property, the integral of which is independent of the path, and use of the symbol "δ" indicates a differential element of some quantity which is not a state function.

$$U = U(V,T) \qquad dU = \left(\frac{\partial U}{\partial V}\right)_T dV + \left(\frac{\partial U}{\partial T}\right)_V dT$$

- ✓ In this manner, we can examine processes in which the volume *V* is maintained constant (***isochore*** or ***isometric*** processes), or the pressure *P* is maintained constant (***isobaric*** processes), or the temperature *T* is maintained constant (***isothermal*** processes). We can also examine ***adiabatic*** processes in which q=0.

Isometric process $\quad dU = \delta q_V \qquad \Delta U = q_V$

Isobaric process

$$H = U + PV$$

$$(U_2 + P_2V_2) - (U_1 + P_1V_1) = H_2 - H_1 = q_P$$

Heat capacity $C = \dfrac{q}{\Delta T}$ $C = \dfrac{\delta q}{\delta T}$

$$C_V = \left(\frac{\delta q}{dT}\right)_V$$ The heat capacity at constant volume

$$C_P = \left(\frac{\delta q}{dT}\right)_P$$ The heat capacity at constant pressure

$$C_V = \left(\frac{\delta q}{dT}\right)_V = \left(\frac{dU'}{dT}\right)_V \qquad dU' = C_V dT$$

$$C_P = \left(\frac{\delta q}{dT}\right)_P = \left(\frac{dH'}{dT}\right)_P \qquad dH' = C_P dT$$

✓ The **specific heat** of the system is the heat capacity per gram at constant *P*, and the **molar heat capacity** is the heat capacity per mole at constant pressure or at constant volume. Thus, for a system containing *n* moles:

$$C_V = nc_V$$

$$C_P = nc_P$$

$$\frac{PdV}{dT} \text{ or } P\left(\frac{\partial V}{\partial T}\right)_P \Rightarrow c_P - c_V = P\left(\frac{\partial V}{\partial T}\right)_P$$

$$H = U + PV$$

$$c_P = \left(\frac{\delta q}{dT}\right)_P = \left(\frac{dH}{dT}\right)_P = \left(\frac{\partial H}{\partial T}\right)_P + P\left(\frac{\partial V}{\partial T}\right)_P$$

$$c_V = \left(\frac{\delta q}{dT}\right)_V = \left(\frac{dU}{dT}\right)_P = \left(\frac{\partial U}{\partial T}\right)_V$$

$$c_P - c_V = \left(\frac{\partial U}{\partial T}\right)_P + P\left(\frac{\partial V}{\partial T}\right)_P - \left(\frac{\partial U}{\partial T}\right)_V$$

$$U = U\left(V,T\right)$$

$$dU = \left(\frac{\partial U}{\partial V}\right)_T dV + \left(\frac{\partial U}{\partial T}\right)_V dT$$

$$\left(\frac{\partial U}{\partial T}\right)_P = \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P + \left(\frac{\partial U}{\partial T}\right)_V$$

$$c_P - c_V = \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P + \left(\frac{\partial U}{\partial T}\right)_V + P\left(\frac{\partial V}{\partial T}\right)_P - \left(\frac{\partial U}{\partial T}\right)_V$$

$$= \left(\frac{\partial V}{\partial T}\right)_P \left[P + \left(\frac{\partial U}{\partial V}\right)_T\right]$$

$$P\left(\frac{\partial V}{\partial T}\right)_P$$

- ✓ The work done by the system per degree rise in temperature in expanding against the constant external pressure *P* acting on the system.

$$\left(\frac{\partial V}{\partial T}\right)_P\left(\frac{\partial U}{\partial V}\right)_T$$

- ✓ The work done per degree rise in temperature in expanding against the internal cohesive forces acting between the constituent particles of the substance.

$$\boxed{dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV}$$

Divided by *dT*

$$\boxed{\frac{dU}{dT} = \left(\frac{\partial U}{\partial T}\right)_V + \left(\frac{\partial U}{\partial V}\right)_T \frac{dV}{dT}}$$

$$\boxed{\left(\frac{\partial U}{\partial T}\right)_P = \left(\frac{\partial U}{\partial T}\right)_V + \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P}$$

Prove: $$\boxed{\left(\frac{\partial U}{\partial T}\right)_P=\left(\frac{\partial U}{\partial T}\right)_V+\left(\frac{\partial U}{\partial V}\right)_T\left(\frac{\partial V}{\partial T}\right)_P}$$

Suppose: $U=U(T,V),\quad V=V(T,P)$

$$dU=\left(\frac{\partial U}{\partial T}\right)_V dT+\left(\frac{\partial U}{\partial V}\right)_T dV$$

$$dV=\left(\frac{\partial V}{\partial T}\right)_P dT+\left(\frac{\partial V}{\partial P}\right)_T dP$$

Substituting the expression of dV, we have:

$$dU=\left(\frac{\partial U}{\partial T}\right)_V dT+\left(\frac{\partial U}{\partial V}\right)_T\left[\left(\frac{\partial V}{\partial T}\right)_P dT+\left(\frac{\partial V}{\partial P}\right)_T dP\right]$$

Rearranging to separate dP from dT, we have:

$$dU = \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial P}\right)_T dP + \left[\left(\frac{\partial U}{\partial T}\right)_V + \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P\right] dT$$

$$= \left(\frac{\partial U}{\partial P}\right)_T dP + \left[\left(\frac{\partial U}{\partial T}\right)_V + \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P\right] dT$$

Because U is a function of T and P. $U = U(T,P)$

$$dU = \left(\frac{\partial U}{\partial P}\right)_T dP + \left(\frac{\partial U}{\partial T}\right)_P dT$$

According to the two expression of dU, we have:

$$\left(\frac{\partial U}{\partial T}\right)_P = \left(\frac{\partial U}{\partial T}\right)_V + \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P$$

✓ For an ideal gas

$$c_P - c_V = R$$

$$\boxed{\left(\frac{\partial U}{\partial V}\right)_T = 0} \quad \boxed{\left(\frac{\partial U}{\partial P}\right)_T = 0} \quad \boxed{\left(\frac{\partial H}{\partial P}\right)_T = 0} \quad \boxed{\left(\frac{\partial H}{\partial V}\right)_T = 0}$$

✓ For a gas

The throttle process is an ***isenthalpic*** process.

✓ For a phase-transformation

$$H_S(T_1)-H_L(T_1)=H_S(T_2)-H_L(T_2)+\int_{T_2}^{T_1}\left(c_P^S-c_P^L\right)dT$$

$$\left(\frac{\partial \Delta H}{\partial T}\right)_P=\Delta c_P$$

✓ Reversible adiabatic processes

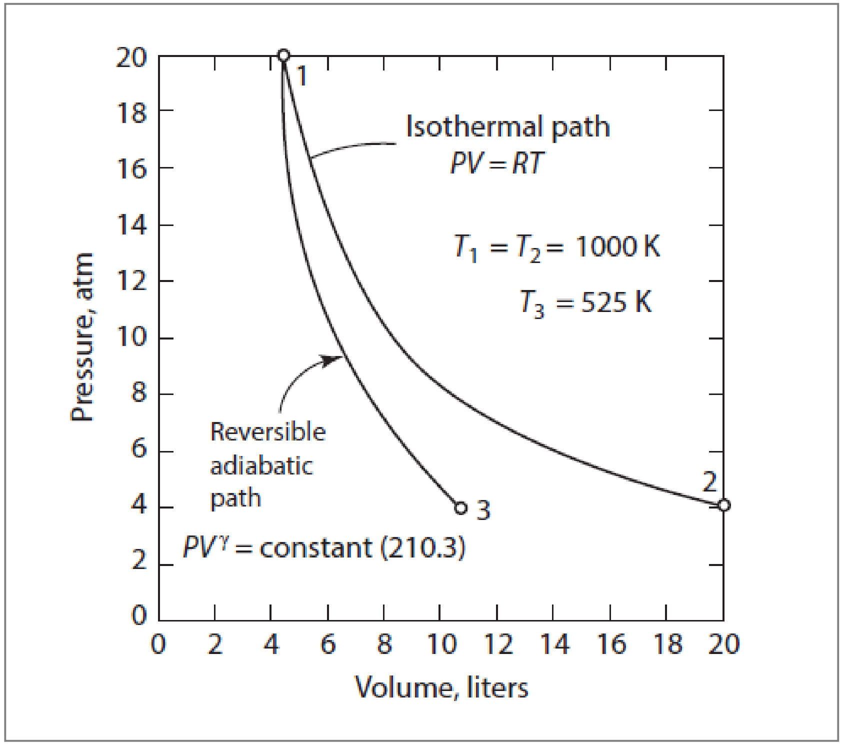
$$dU=c_V dT \qquad \delta w=PdV \qquad C_V dT=-PdV$$

$$P_2V_2^{\gamma}=P_1V_1^{\gamma}=\text{constant}$$

$$\left(\frac{T_2}{T_1}\right)=\left(\frac{V_1}{V_2}\right)^{\gamma-1} \qquad \gamma=\frac{5}{3}$$

✓ Reversible isothermal process

$$w = q = RT \ln \frac{V_2}{V_1} = RT \ln \frac{P_1}{P_2}$$



$$dU = \delta q - \sum \delta w_i$$

✓ Magnetic Work

$$\delta w' = -V\mu_0 \aleph d\mathbf{M}$$

$$\delta w' = -V\mu_0 \int_{M_1}^{M_2} \aleph d\mathbf{M}$$

✓ Electrical Work

$$\delta w' = -V\mathbf{E}d\mathbf{D}$$

$$w' = -V\int_{D_1}^{D_2} \mathbf{E}d\mathbf{D}$$

✓ Surface work

$$\delta w' = -\gamma dA$$

$$\delta w' = -\sigma dA$$

- ✓ A process which involves the spontaneous movement of a system from a non-equilibrium state to an equilibrium state is called **a natural or spontaneous process**.
- ✓ As such a process cannot be reversed without the application of an external agency (a process which would leave a permanent change in the external agency), such a process is said to be irreversible. (The terms natural, spontaneous, and irreversible are synonymous in this context.)

- ✓ If it is considered that an irreversible process is one in which the energy of the system undergoing the process is degraded, then the possibility that the extent of degradation can differ from one process to another suggests that a *quantitative* measure of the extent of degradation, or degree of irreversibility, exists.

- ✓ As the degree of irreversibility of a process is variable, it should be possible for the process to be conducted in such a manner that the degree of irreversibility is minimized. The ultimate of this minimization is a process in which the degree of irreversibility is zero and in which no degradation occurs.

- ✓ This limit, toward which the behavior of real systems can be made to approach, is the reversible process. If a process is reversible, then the concept of spontaneity is no longer applicable.
- ✓ Spontaneity occurred as a result of the system moving, of its own accord, from a non-equilibrium state to an equilibrium state.

- ✓ If the spontaneity is removed, it is apparent that at all times during the process, the system is at equilibrium. Thus **a reversible process** is the one during which the system is never away from equilibrium, and a reversible process which takes the system from the state A to the state B is one in which the process path passes through a continuum of equilibrium states. Such a path is, of course, imaginary.

- ✓ But it is possible to conduct an actual process in such a manner that it is virtually reversible. Such an actual process (***a quasi-static process***) is one which proceeds under the influence of an infinitesimally small driving force such that, during the process, the system is never more than an infinitesimal distance from equilibrium. If, at any point along the path, the external influence is removed, then the process ceases, or, if the direction of the external influence is reversed, then the direction of the process is reversed.

✓ Three points have emerged from the discussion so far:

1. Entropy is a thermodynamic state variable (function).
2. Entropy is not created when a system undergoes a reversible process; entropy is transferred from one part of the system/surroundings to another part.
3. The total entropy of the universe increases when an irreversible process occurs.
4. For all processes, we can write $\Delta S_{system} = q/T + \Delta S_{irr}$ and $q \le q_{rev}$.
5. For all processes, the entropy of the universe increases or stays the same. The total entropy of the universe never decreases.

- ✓ A heat engine is a device which converts heat into work, and it is interesting to note that the first steam engine, which was built in 1769, was in operation for a considerable number of years before the reverse process, i.e., the conversion of work into heat, was investigated.
- ✓ In the operation of a heat engine, some of the energy that has been transferred from a high-temperature heat reservoir is converted into work, with the remainder of the energy being transferred to a low-temperature heat reservoir.

- ✓ The efficiency of a heat engine is given by

$$\text{Efficiency}=\frac{\text{work obtained}}{\text{heat input}}=\frac{w}{q_2}$$

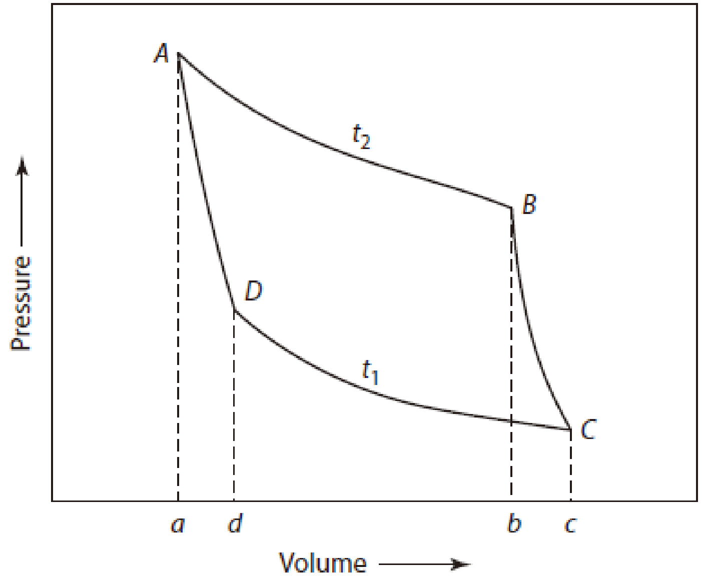


Figure 3.4 A pressure vs. volume depiction of a Carnot cycle.

$$\text{Efficiency}=\frac{w}{q_2}=\frac{q_2-q_1}{q_2}=1-\frac{q_1}{q_2}$$

- The principle of Thomsen states that it is impossible, by means of a cyclic process, to take heat from a reservoir and convert it to work without, in the same operation, transferring heat to a cold reservoir.
- The principle of Clausius states that it is impossible to transfer heat from a cold to a hot reservoir without, in the same process, converting a certain amount of work to heat.

- The efficiency of a Carnot cycle is:

$$\text{Efficiency} = \frac{q_2 - q_1}{q_2} = \frac{T_2 - T_1}{T_2} = 1 - \frac{T_1}{T_2}$$

- This defines an absolute thermodynamic scale of temperature which is independent of the working substance. It is seen that the zero of this temperature scale is that temperature of the cold reservoir which makes the Carnot cycle 100% efficient.

$$dS = \frac{\delta q_{rev}}{T}$$

✓ The Second Law of Thermodynamics can thus be stated as

- The incremental change in entropy dS of a system into which (or out of which) thermal energy has been reversibly added (or expelled) is given as dS=$\delta q_{rev}/T$, and the function S is a state function of the system.

- The entropy of a system in an adiabatic enclosure can never decrease.
  - The entropy increases during an irreversible process.
  - The entropy remains constant during a reversible process.
  - The entropy remains constant if the system was initially in equilibrium.

The Second Law: $$dS_{system} = \frac{\delta q}{T} + dS_{irr}$$

The First Law: $$\delta q = dU_{system} + \delta w$$

$$dS_{system} = \frac{dU_{system} + \delta w}{T} + dS_{irr}$$

$$\delta w = TdS_{system} - dU_{system} - TdS_{irr}$$

$$\delta w \le TdS_{system} - dU_{system}$$

✓ If the temperature remains constant throughout the process (and equal to the temperature of the reservoir supplying heat to the system), then integration from state A to state B gives

$$w \le T\left(S_B - S_A\right) - \left(U_B - U_A\right)$$

$$w_{\max} = q_{rev} = T\left(S_B - S_A\right) - \left(U_B - U_A\right)$$

- ✓ For an incremental change in the state of a closed system, the First Law of Thermodynamics gives:

$$dU = \delta q - \delta w$$

- ✓ If the process occurs reversibly, the Second Law of Thermodynamics gives:

$$dS = \frac{\delta q}{T} \text{ or } \delta q = TdS \qquad \delta w = PdV$$

- ✓ Combination of the two laws gives the equation

$$dU = TdS - PdV$$

- ✓ In an *isolated* system of constant internal energy, U, and constant volume, V, equilibrium is attained when the entropy of the system is a maximum, consistent with the fixed values of U and V.
- ✓ This *spontaneous* process is, by definition, irreversible, and during the movement of the system from its initial non-equilibrium state to its final equilibrium state the entropy of the system increases. The attainment of the equilibrium state coincides with the entropy reaching a *maximum value*, and hence entropy can be used as a criterion for determination of the equilibrium state.

The End